# Real-Data Study of Weekend vs. Weekday Shopping Spending


**Short Summary**

Is there a significant difference between weekend and weekday shopper spending? We show this question is a special case of simple linear regression on a single binary indicator variable, with the regression $t$-statistic proven to coincide exactly with the classical pooled two-sample $t$-statistic under variance homogeneity (Theorem 3.1). Relaxing that assumption, the Behrens–Fisher problem and Welch's approximate solution are derived in full, showing why the pooled and Welch $t$-statistics coincide when $n_1 = n_2$, and how their $p$-values differ (Remark 4.1). These results are applied to a real dataset of 541,909 timestamped UK retail transaction records (Section 5): comparing a realistic $n_1 = n_2 = 500$ sample against the full population it was drawn from shows that the sample can lead us to the wrong conclusion: its $p$-value ($p \approx 0.31$) may lead us to conclude that weekend and weekday shoppers spend similarly, when the population-level $p$-value ($p < 0.05$) shows a real, small difference actually exists. Furthermore, the divergence between the pooled and Welch tests grows with both sample-size imbalance and the genuine possibility of variance heterogeneity in the data.


## Contents

# 1 Introduction

## 1.1 The Motivating Problem

We are interested in shopping habits and whether consumer spending varies between weekends and weekdays. When investigating this through finite samples of shoppers, how can a researcher prove a true difference exists, and how is that assertion quantified? Concretely: suppose $n_1 = 500$ weekend shoppers are sampled and found to spend \$368.86 on average with standard deviation \$382.24, while $n_2 = 500$ weekday shoppers are sampled and found to spend \$458.64 on average with standard deviation \$686.09. Is there sufficient evidence to claim a significant difference in average spending between the two groups, and what is the attained significance level?

We look at two samples drawn from different populations, and we want to know how different these populations actually are, based only on a sample analysis of each. Along the analysis we show that this problem is not a separate technique from linear regression: it *is* a linear regression, in disguise, with a single 0/1 predictor [1]. Seeing the two frameworks as one and the same clarifies exactly what the assumption of variance homogeneity achieves and what breaks when it fails, which is precisely where Welch's test comes in to handle unequal variances.

# 2 The Two-Sample Model

Consider two populations with unknown means $\mu_1, \mu_2$ and unknown variances $\sigma_1^2, \sigma_2^2$. Because neither population is observed directly, inference must proceed from a finite random sample of each. Specializing to our setup, these populations represent the spending amounts of weekend and weekday shoppers, with *each shopper* modeled as an independent, identically distributed normal draw from their respective population:

$$\begin{aligned} Y_{1,1}, \ldots, Y_{1,n_1} &\overset{iid}{\sim} N(\mu_1, \sigma_1^2) \quad \text{(weekend spending)}, \\ Y_{2,1}, \ldots, Y_{2,n_2} &\overset{iid}{\sim} N(\mu_2, \sigma_2^2) \quad \text{(weekday spending)}, \end{aligned} \tag{1}$$

with the two samples mutually independent.[1]

**Proposition 2.1 (Distribution of $\bar{Y}_1 - \bar{Y}_2$, arbitrary variances)**
Define

$$\bar{Y}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} Y_{1,i}, \qquad \bar{Y}_2 = \frac{1}{n_2}\sum_{j=1}^{n_2} Y_{2,j}$$

for the sample means of the two groups. We then have:

$$\bar{Y}_1 - \bar{Y}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right). \tag{2}$$

*Proof.*

Since $Y_{1,1}, \ldots, Y_{1,n_1}$ are i.i.d. $N(\mu_1, \sigma_1^2)$, a linear combination of independent normals is itself normal, with mean and variance given by:

$$\bar{Y}_1 = \sum_{i=1}^{n_1} \frac{1}{n_1} Y_{1,i} \quad \Longrightarrow \quad \mathrm{E}[\bar{Y}_1] = \sum_{i=1}^{n_1} \frac{1}{n_1}\mu_1 = \mu_1, \qquad \mathrm{Var}(\bar{Y}_1) = \sum_{i=1}^{n_1} \frac{1}{n_1^2}\sigma_1^2 = \frac{\sigma_1^2}{n_1}.$$

[1] Formally, mutual independence implies that every pair of observations across samples has zero covariance, $\mathrm{Cov}(Y_{1,i}, Y_{2,j}) = 0$ for all $i$ and $j$, allowing the joint probability density to factorize as $f_{\mathbf{Y}_1,\mathbf{Y}_2}(\mathbf{y}_1, \mathbf{y}_2) = \prod_{i=1}^{n_1} f_{Y_{1,i}}(y_{1,i}) \prod_{j=1}^{n_2} f_{Y_{2,j}}(y_{2,j})$.

This gives
$$\bar{Y}_1 \sim N(\mu_1, \sigma_1^2/n_1), \qquad \bar{Y}_2 \sim N(\mu_2, \sigma_2^2/n_2).$$

Since the two samples are independent, $\bar{Y}_1$ and $\bar{Y}_2$ are independent. A difference of independent normal random variables is again normal, with mean and variance given by:

$$\mathrm{E}[\bar{Y}_1 - \bar{Y}_2] = \mu_1 - \mu_2, \qquad \mathrm{Var}(\bar{Y}_1 - \bar{Y}_2) = \mathrm{Var}(\bar{Y}_1) + \mathrm{Var}(\bar{Y}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

This gives $\bar{Y}_1 - \bar{Y}_2 \sim N\big(\mu_1 - \mu_2,\ \sigma_1^2/n_1 + \sigma_2^2/n_2\big)$. □

Standardizing (2) gives

$$Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} \sim N(0,1). \tag{3}$$

Testing $H_0 : \mu_1 = \mu_2$ with $Z$ is not possible in practice: $Z$ depends on $\sigma_1^2, \sigma_2^2$, and these are unknown. The solution is to replace $\sigma_1^2, \sigma_2^2$ with the sample variances, given by:[2]

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (Y_{1,i} - \bar{Y}_1)^2, \qquad S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (Y_{2,j} - \bar{Y}_2)^2.$$

**Definition 2.1 (Variance homogeneity)**
The samples in (1) satisfy *variance homogeneity* if

$$\sigma_1^2 = \sigma_2^2 = \sigma^2 \tag{4}$$

for some common, unknown $\sigma^2$.

**Theorem 2.1 (Pooled two-sample *t*-statistic)**
Under the two-sample setup (1) with variance homogeneity (4), when testing the null hypothesis $H_0 : \mu_1 = \mu_2$ against the two-sided alternative hypothesis $H_a : \mu_1 \neq \mu_2$, the test statistic

$$T = \frac{\bar{Y}_1 - \bar{Y}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \tag{5}$$

follows a Student $t$-distribution with $n_1 + n_2 - 2$ degrees of freedom under $H_0$, where $S_p^2$ denotes the pooled sample variance:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

*Proof.*

Specializing Proposition 2.1 to $\sigma_1^2 = \sigma_2^2 = \sigma^2$,

$$\bar{Y}_1 - \bar{Y}_2 \sim N\left(\mu_1 - \mu_2,\ \sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right).$$

Standardizing this normal random variable gives

$$Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sigma \sqrt{1/n_1 + 1/n_2}} \sim N(0,1).$$

[2] These are unbiased estimators of the population variances: $\mathrm{E}[S_1^2] = \sigma_1^2$ and $\mathrm{E}[S_2^2] = \sigma_2^2$. The $n_i - 1$ divisor ensures unbiasedness: dividing by $n_i$ instead would systematically underestimate $\sigma_i^2$, since the sample mean is itself estimated from the data, using up one degree of freedom.

Because both group variances estimate the same parameter $\sigma^2$, combining them into a single pooled estimate $S_p^2$ incorporates all $n_1 + n_2 - 2$ degrees of freedom from both samples. From normal-sample theory, each group's rescaled sample variance is individually chi-square distributed (see Appendix A):

$$\frac{(n_1-1)S_1^2}{\sigma^2} \sim \chi^2_{n_1-1}, \qquad \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{n_2-1},$$

independently of one another. A sum of independent chi-square random variables is again chi-square, with degrees of freedom adding up:

$$W := \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} = \frac{(n_1+n_2-2)S_p^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}.$$

Without a common $\sigma^2$ to divide through by, $(n_1-1)S_1^2/\sigma_1^2$ and $(n_2-1)S_2^2/\sigma_2^2$ are each individually chi-square, but their sum is not proportional to any chi-square distribution once $\sigma_1^2 \neq \sigma_2^2$.

Since the sample means $(\bar{Y}_1, \bar{Y}_2)$ and sample variances $(S_1^2, S_2^2)$ are independent for normal data, $Z$ and $W$ are independent (see Appendix A). By the definition of the $t$-distribution, if $Z \sim N(0,1)$ and $W \sim \chi^2_k$ are independent, then (see Appendix B):

$$T = \frac{Z}{\sqrt{W/k}} \sim t_k,$$

where $T$ is Student's $t$-statistic. Substituting $Z$ and $W$ above, with $k = n_1 + n_2 - 2$,

$$T = \frac{\dfrac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}}}{\sqrt{S_p^2/\sigma^2}}.$$

The two factors of $\sigma$ cancel exactly, leaving

$$T = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2}.$$

Under $H_0 : \mu_1 = \mu_2$, the difference $(\mu_1 - \mu_2)$ in the numerator equals zero, yielding the test statistic in (5). Under the null hypothesis $H_0$, $T_{n_1+n_2-2}$ is the random variable whose distribution is the single observed value computed from the actual data. The $p$-value quantifies, under the assumption that $H_0$ holds, the probability of observing a test statistic at least as extreme as the one obtained. For the two-sided alternative hypothesis $H_a : \mu_1 \neq \mu_2$, the relevant event is $|T_{n_1+n_2-2}| > |t_{\text{obs}}|$:

$$p = P(|T_{n_1+n_2-2}| > |t_{\text{obs}}|)\,.$$

This event splits into two disjoint tails, $T_{n_1+n_2-2} > |t_{\text{obs}}|$ or $T_{n_1+n_2-2} < -|t_{\text{obs}}|$, so

$$p = P(T_{n_1+n_2-2} > |t_{\text{obs}}|) + P(T_{n_1+n_2-2} < -|t_{\text{obs}}|)\,.$$

Since the $t$-distribution is symmetric about 0, both tails have equal probability, $P(T_{n_1+n_2-2} < -|t_{\text{obs}}|) = P(T_{n_1+n_2-2} > |t_{\text{obs}}|)$, so

$$p = 2P(T_{n_1+n_2-2} > |t_{\text{obs}}|)\,. \qquad \square$$

# 3 Reformulating as Linear Regression

## 3.1 The Dummy-Variable Model

Let $n = n_1 + n_2$ denote the total number of observations, where $n_1$ represents weekend shoppers and $n_2$ represents weekday shoppers. Define the binary group indicator variable $D_i$ for each observation $i = 1, \ldots, n$ as

$$D_i = \begin{cases} 1 & \text{if observation } i \text{ is a weekend shopper,} \\ 0 & \text{if observation } i \text{ is a weekday shopper.} \end{cases}$$

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 D_i + \varepsilon_i, \qquad i = 1, \ldots, n. \tag{6}$$

Expressing (6) in matrix notation yields $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where

$$\mathbf{y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & D_1 \\ \vdots & \vdots \\ 1 & D_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

Under standard classical linear regression assumptions, the Ordinary Least Squares (OLS) estimator of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \tag{7}$$

This formulation represents a standard linear regression model containing an intercept (constant term) and a single binary indicator covariate.

**Theorem 3.1 (Equivalence of Regression and Two-Sample $t$-Test)** Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be the OLS estimators given by (7) for model (6). Then

$$\hat{\beta}_0 = \bar{Y}_2, \qquad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_2.$$

Furthermore, the residual sum of squares (RSS) equals the pooled sum of squares,

$$\text{RSS} = (n_1 - 1)S_1^2 + (n_2 - 1)S_2^2,$$

and under the assumption of homoscedastic errors, the standard test statistic for testing $H_0 : \beta_1 = 0$ is algebraically identical to the pooled two-sample $t$-statistic given in (5), with $n - 2 = n_1 + n_2 - 2$ degrees of freedom.

The full derivation of Theorem 3.1 is provided in Appendix C. Conceptually, the intercept $\hat{\beta}_0$ represents the mean of the reference group ($\bar{Y}_2$, weekday spending), while the slope $\hat{\beta}_1$ measures the difference between the weekend mean and the weekday mean ($\bar{Y}_1 - \bar{Y}_2$).

Consequently, testing the null hypothesis $H_0 : \beta_1 = 0$ in the regression model is mathematically equivalent to testing $H_0 : \mu_1 - \mu_2 = 0$ in the two-sample framework 2. Because both methods evaluate the same underlying hypothesis, they yield the exact same test statistic $t_{\text{obs}}$ with $n_1 + n_2 - 2$ degrees of freedom, and consequently the same two-sided $p$-value:

$$p = 2\mathbb{P}(T_{n_1+n_2-2} > |t_{\text{obs}}|).$$

# 4 Variance Heterogeneity: When Homogeneity Fails

When comparing weekend and weekday shopping, real-world spending data rarely satisfies the assumption of equal variances ($\sigma_1^2 = \sigma_2^2$). As a result, the theoretical derivation of the pooled $t$-test (Theorem 2.1) breaks down, requiring a different set of tools.

### 4.1 The Behrens–Fisher Problem

Without a shared variance, $S_1^2$ and $S_2^2$ estimate two distinct population parameters, $\sigma_1^2$ and $\sigma_2^2$. Consequently, they cannot be combined into a single chi-square-distributed sum. More precisely, while the difference in sample means remains normally distributed:

$$\bar{Y}_1 - \bar{Y}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right),$$

the unpooled variance estimator $U = S_1^2/n_1 + S_2^2/n_2$ fails to follow a standard chi-square distribution. Expressing $U$ in terms of independent chi-square variables $V_1 \sim \chi^2_{n_1-1}$ and $V_2 \sim \chi^2_{n_2-1}$ reveals why:

$$U = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} = \frac{\sigma_1^2}{n_1(n_1-1)}V_1 + \frac{\sigma_2^2}{n_2(n_2-1)}V_2.$$

Because $U$ is a linear combination of *differently scaled* chi-square random variables, $U$ itself is not proportional to any single chi-square distribution unless $\sigma_1^2 = \sigma_2^2$.

As a result, the unpooled test statistic:

$$T' = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \tag{8}$$

has no exact, closed-form Student $t$-distribution for any finite sample size.[3]

### 4.2 Welch Approximation

Welch's method resolves this problem by approximating the distribution of the unpooled sample variance $U = \frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}$ with a scaled chi-square variable $cW$, where $W \sim \chi^2_\nu$, and $\nu$ is the effective degrees of freedom (which need not be an integer).

Because $S_1^2$ and $S_2^2$ are unbiased estimators of $\sigma_1^2$ and $\sigma_2^2$, the expected value of $U$ gives the true variance of the difference between sample means:

$$\mathbb{E}[U] = \mathrm{Var}(\bar{Y}_1 - \bar{Y}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \equiv \sigma_\Delta^2.$$

To expose the standard Student $t$-ratio structure, namely a standard normal variable divided by the square root of a normalized variance, we divide both the numerator and denominator of $T'$ by the true standard deviation $\sigma_\Delta = \sqrt{\mathbb{E}[U]}$:

$$T' = \frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)}{\sqrt{U}} = \frac{\dfrac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sigma_\Delta}}{\sqrt{\dfrac{U}{\sigma_\Delta^2}}} = \frac{\dfrac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sqrt{\mathbb{E}[U]}}}{\sqrt{\dfrac{U}{\mathbb{E}[U]}}}.$$

This algebraic division explicitly separates $T'$ into two independent components:

1. **Standard Normal Numerator:** Because $(\bar{Y}_1 - \bar{Y}_2)$ is normal with mean $\mu_1 - \mu_2$ and variance $\sigma_\Delta^2 = \mathrm{E}[U]$, the numerator is exactly a standard normal random variable:

$$Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sqrt{\mathrm{E}[U]}} \sim N(0, 1).$$

[3] The *null distribution* of a test statistic is its distribution under $H_0$, which dictates $p$-value computation. For the pooled $t$-statistic in Theorem 2.1, the exact null distribution is $t_{n_1+n_2-2}$. For $T'$, no exact named distribution exists; this analytical challenge is known as the *Behrens–Fisher problem.*

2. **Approximate Chi-Square Denominator:** Welch's approximation models $U \approx cW'$ where $W' \sim \chi^2_\nu$. Equating expectations gives $\mathbb{E}[U] = \mathbb{E}[cW'] = c\nu$, which simplifies the normalized denominator term:
$$\frac{U}{\mathbb{E}[U]} \approx \frac{cW'}{c\nu} = \frac{W'}{\nu}.$$

Notice that the scaling constant $c$ and the unknown population variance $\sigma^2_\Delta$ cancel out completely. Substituting $Z$ and $W'/\nu$ back into $T'$ yields the classical definition of a Student $t$ random variable:
$$T' \approx \frac{Z}{\sqrt{W'/\nu}} \sim t_\nu.$$
Thus, approximating $U$ as a scaled chi-square variable shows that $T'$ follows an approximate Student $t$-distribution with $\nu$ degrees of freedom. The parameter $\nu$ is determined by matching the first and second moments of $cW'$ to those of $U$.

For normal samples, the sample variances satisfy $\mathrm{E}[S_i^2] = \sigma_i^2$ and $\mathrm{Var}(S_i^2) = \frac{2\sigma_i^4}{n_i-1}$,[4] which yields:
$$\mathrm{E}[U] = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}, \qquad \mathrm{Var}(U) = \frac{2}{n_1-1}\left(\frac{\sigma_1^2}{n_1}\right)^2 + \frac{2}{n_2-1}\left(\frac{\sigma_2^2}{n_2}\right)^2. \tag{9}$$

Equating the mean and variance of $cW'$ ($\mathrm{E}[cW'] = c\nu$, $\mathrm{Var}(cW') = 2c^2\nu$) to $\mathrm{E}[U]$ and $\mathrm{Var}(U)$ produces the system:
$$c\nu = \mathrm{E}[U] \qquad \text{and} \qquad 2c^2\nu = \mathrm{Var}(U).$$
This gives:
$$\nu = \frac{2(\mathrm{E}[U])^2}{\mathrm{Var}(U)}. \tag{10}$$
Substituting (9) into (10) yields:
$$\nu = \frac{\left(S_1^2/n_1 + S_2^2/n_2\right)^2}{\dfrac{(S_1^2/n_1)^2}{n_1-1} + \dfrac{(S_2^2/n_2)^2}{n_2-1}}. \tag{11}$$
Under this approximation, the test statistic follows an approximate Student $t$-distribution with $\nu$ degrees of freedom ($T' \sim t_\nu$), which constitutes Welch's two-sample $t$-test.

*Remark 4.1 (Balanced Designs: Coincidence of Pooled and Welch Test Statistics)*
When sample sizes are equal ($n_1 = n_2 = n$), the pooled standard error $\mathrm{SE}_{\text{pooled}} = S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ and the Welch standard error $\mathrm{SE}_{\text{Welch}} = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$ are algebraically identical.

To see this explicitly, recall from Theorem 2.1 that the pooled variance is $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$. When $n_1 = n_2 = n$, this simplifies to the average of the two sample variances:
$$S_p^2 = \frac{(n-1)S_1^2 + (n-1)S_2^2}{2n-2} = \frac{S_1^2 + S_2^2}{2}.$$
Substituting this $S_p^2$ into the squared pooled standard error yields:
$$\mathrm{SE}^2_{\text{pooled}} = S_p^2\left(\frac{1}{n} + \frac{1}{n}\right) = \left(\frac{S_1^2 + S_2^2}{2}\right)\left(\frac{2}{n}\right) = \frac{S_1^2}{n} + \frac{S_2^2}{n} = \mathrm{SE}^2_{\text{Welch}}.$$

[4] By Theorem A.1, $V_i = \frac{(n_i-1)S_i^2}{\sigma_i^2} \sim \chi^2_{n_i-1}$. Because a chi-square random variable with $k$ degrees of freedom has mean $k$ and variance $2k$, we have $\mathrm{E}[V_i] = n_i - 1$ and $\mathrm{Var}(V_i) = 2(n_i-1)$. Rescaling by $\frac{\sigma_i^2}{n_i-1}$ yields $\mathrm{E}[S_i^2] = \sigma_i^2$ and $\mathrm{Var}(S_i^2) = \left(\frac{\sigma_i^2}{n_i-1}\right)^2 \cdot 2(n_i-1) = \frac{2\sigma_i^4}{n_i-1}$.

Taking the square root gives $\mathrm{SE}_{\text{pooled}} = \mathrm{SE}_{\text{Welch}}$. The pooled test statistic $T$ in (5) and the Welch test statistic $T'$ in (8) take the exact same numerical value ($T = T'$), because their numerators $(\bar{Y}_1 - \bar{Y}_2)$ and standard errors are identical.

Consequently, when sample sizes are equal ($n_1 = n_2 = n$), any discrepancy between the two tests arises solely from their reference degrees of freedom. Substituting $n_1 = n_2 = n$ into (11) yields the ratio between Welch's degrees of freedom ($\nu$) and the pooled degrees of freedom ($n_1 + n_2 - 2 = 2n - 2$):

$$\frac{\nu}{n_1 + n_2 - 2} = \frac{\nu}{2n - 2} = \frac{(S_1^2 + S_2^2)^2}{2(S_1^4 + S_2^4)},$$

a ratio that depends exclusively on the sample variance ratio $S_1^2/S_2^2$, independent of the sample size $n$.

Equal test statistics do not imply identical $p$-values. Both tests evaluate $p = 2\mathbb{P}(T > |t_{\text{obs}}|)$, but against distinct reference distributions: the pooled test evaluates $T \sim t_{n_1+n_2-2}$, whereas Welch's test evaluates $T \sim t_\nu$. Because

$$\nu \leq n_1 + n_2 - 2, \qquad \text{with equality holding if and only if } S_1^2 = S_2^2,$$ [5]

Welch's reference distribution has fewer degrees of freedom and heavier tails. Evaluating $t_{\text{obs}}$ against a heavier-tailed distribution assigns greater probability to the tails, producing a larger $p$-value whenever sample variances differ. This matches the empirical results in Section 5, where $t = 0.3777$ yields $p = 0.7077$ under the pooled test versus $p = 0.7079$ under Welch's test.

# 5 Empirical Validation on Real Data

We apply this model to the study of weekend versus weekday shopper spending, using the UCI "Online Retail" dataset (Chen, 2015), which contains 541,909 real, timestamped transaction records from a UK online retailer between December 2010 and December 2011.

## 5.1 A Realistic Sample Against the Full Population

We evaluate both a realistic sample ($n_1 = n_2 = 500$) and the full population from which it was drawn across both raw and trimmed stages of the data. Here, trimming refers to standard outlier removal: writing $Q_1$ and $Q_3$ for the first and third quartiles of the combined dataset, any purchase order exceeding $Q_3 + 1.5(Q_3 - Q_1)$ is excluded using a common cutoff across both groups.

| Stage | $n_1/n_2$ | mean gap | $p$ (pooled) | $p$ (Welch) |
|---|---|---|---|---|
| Raw sample | 500/500 | \$89.78 | 0.01073 | 0.01077 |
| Trimmed sample | 474/444 | \$13.58 | 0.3067 | 0.3072 |
| Full population (untrimmed) | 2,204/17,755 | \$185.06 | $4.08 \times 10^{-6}$ | $3.26 \times 10^{-24}$ |
| **Full population (trimmed)** | 2,117/16,032 | \$10.07 | **0.0454** | **0.0337** |

In the table above, the *mean gap* denotes the observed absolute difference between average weekend and weekday spending, $|\bar{Y}_1 - \bar{Y}_2|$.

*Remark 5.1 (A non-significant sample result is not evidence of no difference)*
A researcher working only with the trimmed sample (474/444) would compute $p \approx 0.31$ and could reasonably, but incorrectly, conclude "no evidence of a difference, weekend and weekday shoppers are essentially the same." At the full-population scale, the exact same trimming

[5] For $n_1 = n_2$, expanding $(S_1^2+S_2^2)^2 \leq 2(S_1^4+S_2^4)$ yields $2S_1^2S_2^2 \leq S_1^4+S_2^4$, which is equivalent to $0 \leq (S_1^2-S_2^2)^2$. Equality holds if and only if $S_1^2 = S_2^2$.

procedure still yields $p < 0.05$ by both tests (pooled 0.0454, Welch 0.0337): a real difference does exist. The correct conclusion from the sample alone should have been "not enough evidence to detect a difference," not "no difference exists".

*Remark 5.2 (Pooled/Welch divergence tracks both imbalance and heteroscedasticity)*
The table traces Remark 4.1 across varying levels of sample imbalance. At $n_1 = n_2 = 500$, the $t$-statistics coincide exactly ($t = -2.5561$), differing only in degrees of freedom (998 vs. 781.55). At $n_1 = 474$ and $n_2 = 444$, the test statistics remain nearly identical ($t = 1.0227$ vs. 1.0218). At the population level ($n_1 \approx 2{,}200$ vs. $n_2 \approx 18{,}000$), the two tests yield noticeably different $p$-values (0.0454 vs. 0.0337), signaling that the assumption of **equal variance across weekend and weekday spending may fail.**

## 6 Conclusion

The two-sample $t$-test is a special case of linear regression with a single binary predictor. Assuming equal variances across groups matches the classical OLS variance homogeneity assumption, which yields the standard Student $t$-distribution (Theorem 2.1). When group variances differ, Welch's test replaces this assumption with data-dependent degrees of freedom to approximate the null distribution. In balanced designs ($n_1 = n_2$), the pooled and Welch test statistics take the same numerical value, but their $p$-values still differ whenever sample variances are unequal because the two tests evaluate that statistic against different degrees of freedom (Remark 4.1).

Applied to 541,909 retail transaction records, a sample of 500 observations per group illustrates two key risks. First, a non-significant sample result can reflect a limited sample size rather than the absence of a real difference (Remark 5.1). Second, divergence between pooled and Welch test results grows with sample imbalance, serving as a sign that equal variance assumptions between samples may fail (Remark 5.2).

# Appendices

## Appendix A Appendix A: Distributional Theory for Normal Random Samples

### A.1 Fundamental Chi Square Properties

**Definition A.1 (Chi Square Distribution)**
Let $Z_1, \dots, Z_k$ be independent and identically distributed standard normal random variables,

$$Z_1, \dots, Z_k \overset{\text{iid}}{\sim} N(0, 1).$$

The distribution of the random variable

$$Q = \sum_{i=1}^{k} Z_i^2$$

is called the chi square distribution with $k$ degrees of freedom, denoted $\chi_k^2$.

**Lemma A.1 (Additivity of Independent Chi Square Variables)**
Let $U \sim \chi_{k_1}^2$ and $V \sim \chi_{k_2}^2$ be independent random variables. Then

$$U + V \sim \chi_{k_1+k_2}^2.$$

*Proof.*

By definition, since $U$ and $V$ are chi square distributed, there exist i.i.d. standard normal random variables $Z_1, \dots, Z_{k_1}$ and $W_1, \dots, W_{k_2}$ such that

$$U = \sum_{i=1}^{k_1} Z_i^2 \qquad \text{and} \qquad V = \sum_{j=1}^{k_2} W_j^2.$$

The independence of $U$ and $V$ allows all $k_1 + k_2$ random variables $Z_1, \dots, Z_{k_1}, W_1, \dots, W_{k_2}$ to be taken mutually independent. Setting $(T_1, \dots, T_{k_1+k_2}) := (Z_1, \dots, Z_{k_1}, W_1, \dots, W_{k_2})$, these are i.i.d. $N(0, 1)$, and

$$U + V = \sum_{i=1}^{k_1} Z_i^2 + \sum_{j=1}^{k_2} W_j^2 = \sum_{\ell=1}^{k_1+k_2} T_\ell^2.$$

By definition of the chi square distribution, $U + V \sim \chi_{k_1+k_2}^2$. □

### A.2 Independence and Distribution of Sample Statistics

**Theorem A.1 (Sampling Distribution of Sample Mean and Variance)**
Let $Y_1, \dots, Y_n$ be i.i.d. $N(\mu, \sigma^2)$, and define the sample mean and sample variance as

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^{n} Y_i \qquad \text{and} \qquad S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \bar{Y})^2.$$

Then $\bar{Y}$ and $S^2$ are independent, with sampling distributions:

$$\bar{Y} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \qquad \text{and} \qquad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

*Proof.*

We first establish the sampling distribution of $\bar{Y}$. Since $Y_1, \ldots, Y_n$ are independent normal random variables, any linear combination is also normally distributed. Calculating the expectation and variance yields:

$$\mathbb{E}[\bar{Y}] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[Y_i] = \mu \qquad \text{and} \qquad \operatorname{Var}(\bar{Y}) = \frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}(Y_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

Thus, $\bar{Y} \sim N(\mu, \sigma^2/n)$, which standardizes to $Z_{\bar{Y}} = \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$.

To prove independence of $\bar{Y}$ and $S^2$ alongside the chi square distribution of $S^2$, standardize the data by setting $Z_i = (Y_i - \mu)/\sigma$ for $i = 1, \ldots, n$. Then $\mathbf{Z} = (Z_1, \ldots, Z_n)^\top \sim N(\mathbf{0}, \mathbf{I}_n)$, and the standardized sample mean is:

$$\bar{Z} = \frac{1}{n}\sum_{i=1}^{n} Z_i = \frac{\bar{Y}-\mu}{\sigma}.$$

Consider the unit vector in $\mathbb{R}^n$:

$$\mathbf{o}_1 = \left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \ldots, \frac{1}{\sqrt{n}}\right).$$

Since $\|\mathbf{o}_1\| = \sqrt{n \cdot (1/n)} = 1$, $\mathbf{o}_1$ forms a valid one dimensional orthonormal basis for $\operatorname{span}(\mathbf{o}_1)$. By the Gram Schmidt process, we can choose $n-1$ additional unit vectors $\mathbf{o}_2, \ldots, \mathbf{o}_n \in \mathbb{R}^n$ that are mutually orthogonal to each other and to $\mathbf{o}_1$. Stacking these rows yields an $n \times n$ orthogonal matrix $\mathbf{O}$ satisfying $\mathbf{O}\mathbf{O}^\top = \mathbf{O}^\top\mathbf{O} = \mathbf{I}_n$.

Define the rotated random vector $\mathbf{V} = (V_1, \ldots, V_n)^\top$ by $\mathbf{V} = \mathbf{O}\mathbf{Z}$. The multivariate normal $\mathbf{Z}$ transforms as follows:

$$\mathbf{V} = \mathbf{O}\mathbf{Z} \sim N\Big(\mathbf{O}\mathbf{0}, \mathbf{O}\mathbf{I}_n\mathbf{O}^\top\Big) = N(\mathbf{0}, \mathbf{I}_n).$$

Hence, $V_1, \ldots, V_n$ are independent and identically distributed $N(0,1)$ random variables. By design of the first row $\mathbf{o}_1$:

$$V_1 = \mathbf{o}_1^\top\mathbf{Z} = \sum_{i=1}^{n}\frac{1}{\sqrt{n}}Z_i = \sqrt{n}\,\bar{Z} = \sqrt{n}\left(\frac{\bar{Y}-\mu}{\sigma}\right).$$

Because orthogonal transformations preserve vector lengths ($\|\mathbf{V}\|^2 = \|\mathbf{Z}\|^2$), the total sum of squares decomposes as:

$$\sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n} V_i^2.$$

Expanding the sample variance numerator in terms of $Z_i$:

$$\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n}(Z_i - \bar{Z})^2 = \sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2 = \sum_{i=1}^{n} V_i^2 - V_1^2 = \sum_{i=2}^{n} V_i^2.$$

We now establish independence and the remaining distribution:

1. **Independence**: The components $V_1, V_2, \ldots, V_n$ are mutually independent. The sample mean $\bar{Y} = \mu + \frac{\sigma}{\sqrt{n}}V_1$ is only a function of $V_1$. The sample variance $S^2 = \frac{\sigma^2}{n-1}\sum_{i=2}^{n} V_i^2$ is a function of $(V_2, \ldots, V_n)$. Because functions of disjoint sets of independent random variables are independent, $\bar{Y}$ and $S^2$ are independent.

2. **Chi square distribution**: The sum $\sum_{i=2}^{n} V_i^2$ consists of $n-1$ independent squared standard normal variables $V_2, \ldots, V_n \overset{\text{iid}}{\sim} N(0,1)$. By definition of the chi square distribution:

$$\frac{(n-1)S^2}{\sigma^2} = \sum_{i=2}^{n} V_i^2 \sim \chi^2_{n-1}. \qquad \square$$

# Appendix B Appendix B: Exact Sampling Theory for Student $t$ Statistics

## B.1 Ratio Representation and Density Derivation

**Definition B.1 (Student $t$ Distribution)**
A random variable $T$ has the Student $t$ distribution with $k > 0$ degrees of freedom, written $T \sim t_k$, if its probability density function is given by

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{t^2}{k}\right)^{-\frac{k+1}{2}}, \qquad t \in \mathbb{R},$$

where $\Gamma(\cdot)$ denotes the Gamma function.

**Theorem B.1 (Ratio Representation of Student $t$ Distribution)**
Let $Z \sim N(0,1)$ and $W \sim \chi^2_k$ be independent random variables. Then

$$T := \frac{Z}{\sqrt{W/k}} \sim t_k.$$

*Proof.*

The probability density function of $Z \sim N(0,1)$ is $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, and the density of $W \sim \chi^2_k$ is given by

$$f_W(w) = \frac{1}{2^{k/2}\Gamma(k/2)} w^{k/2-1} e^{-w/2}, \qquad w > 0.$$

Since $Z$ and $W$ are independent, their joint density factorizes as $f_{Z,W}(z,w) = \varphi(z) f_W(w)$.

We transform $(Z,W)$ to $(T,W)$ via $T = \frac{Z}{\sqrt{W/k}}$ and $W = W$. The inverse transformation is:

$$z(t,w) = t\sqrt{\frac{w}{k}} \qquad \text{and} \qquad w(t,w) = w.$$

The $2 \times 2$ Jacobian matrix of the inverse transformation is:

$$\mathbf{J} = \begin{pmatrix} \dfrac{\partial z}{\partial t} & \dfrac{\partial z}{\partial w} \\ \dfrac{\partial w}{\partial t} & \dfrac{\partial w}{\partial w} \end{pmatrix} = \begin{pmatrix} \sqrt{\dfrac{w}{k}} & \dfrac{t}{2\sqrt{kw}} \\ 0 & 1 \end{pmatrix},$$

with determinant $|\det \mathbf{J}| = \sqrt{\frac{w}{k}}$. The joint density of $(T,W)$ is therefore:

$$f_{T,W}(t,w) = f_{Z,W}\big(z(t,w), w\big) \cdot |\det \mathbf{J}| = \varphi\left(t\sqrt{\frac{w}{k}}\right) f_W(w) \sqrt{\frac{w}{k}}.$$

Integrating over $w \in (0,\infty)$ yields the marginal density of $T$:

$$f_T(t) = \int_0^\infty \varphi\left(t\sqrt{w/k}\right) f_W(w) \sqrt{w/k}\, dw = \frac{1}{\sqrt{2\pi k}\, 2^{k/2}\Gamma(k/2)} \int_0^\infty w^{(k-1)/2} \exp\left(-\frac{w}{2}\left(1 + \frac{t^2}{k}\right)\right) dw.$$

Define $a = \frac{1}{2}\left(1 + \frac{t^2}{k}\right)$ and $s = \frac{k+1}{2}$. Applying the standard Gamma integral identity $\int_0^\infty w^{s-1} e^{-aw}\, dw = \frac{\Gamma(s)}{a^s}$, the integral evaluates to:

$$\int_0^\infty w^{(k-1)/2} e^{-aw}\, dw = \frac{\Gamma\left(\frac{k+1}{2}\right)}{a^{(k+1)/2}}.$$

Substituting this result back into $f_T(t)$ yields:

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{2\pi k}\,2^{k/2}\Gamma(k/2)}\left(\frac{2}{1+t^2/k}\right)^{\frac{k+1}{2}} = \frac{\Gamma\left(\frac{k+1}{2}\right)2^{1/2}}{\sqrt{2\pi k}\,\Gamma(k/2)}\left(1+\frac{t^2}{k}\right)^{-\frac{k+1}{2}}.$$

Since $\frac{2^{1/2}}{\sqrt{2\pi k}} = \frac{1}{\sqrt{k\pi}}$, this simplifies directly to

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)}\left(1+\frac{t^2}{k}\right)^{-\frac{k+1}{2}},$$

which matches the density of $t_k$, completing the proof. □

## B.2 Exact Application to Two Sample Pooled Testing

*Remark B.1 (Role of the Independence Assumption)*
Theorem B.1 strictly relies on the independence of $Z$ and $W$. The variable transformation and marginal integration rest directly on $f_{Z,W}(z,w) = \varphi(z)f_W(w)$. In our setup, weekend spending $Y_1 \sim N(\mu_1, \sigma^2)$ and weekday spending $Y_2 \sim N(\mu_2, \sigma^2)$ are modeled as independent normal populations. Theorem A.1 (Appendix A) guarantees that for each population, the sample mean and sample variance are mutually independent: $\bar{Y}_1 \sim N(\mu_1, \sigma^2/n_1)$ is independent of $S_1^2$, and $\bar{Y}_2 \sim N(\mu_2, \sigma^2/n_2)$ is independent of $S_2^2$.

Because weekend and weekday spending samples are mutually independent, the mean difference $(\bar{Y}_1 - \bar{Y}_2)$ is normally distributed with mean $\mu_1 - \mu_2$ and variance $\sigma^2(1/n_1 + 1/n_2)$. Standardizing this difference yields the standard normal random variable:

$$Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)}{\sigma\sqrt{1/n_1 + 1/n_2}} \sim N(0,1).$$

Furthermore, the scaled pooled sample variance:

$$W = \frac{(n_1+n_2-2)S_p^2}{\sigma^2} = \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}$$

is a chi square random variable by additivity of the independent scaled sample variances from both shopper groups. Because $(\bar{Y}_1, \bar{Y}_2)$ are mutually independent of $(S_1^2, S_2^2)$, $Z$ depends strictly on the group sample means while $W$ depends strictly on the group sample variances. Thus, $Z$ and $W$ form an independent $(N(0,1), \chi^2_{n_1+n_2-2})$ pair, satisfying the exact hypothesis required by Theorem 2.1.

# Appendix C Appendix C: General Least Squares Theory and Regression Equivalence

## C.1 Matrix Algebra of Normal Equations

The ordinary least squares (OLS) estimator minimizes the sum of squared residuals:

$$Q(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top\mathbf{y} - 2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y} + \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}.$$

Setting the gradient $\nabla_{\boldsymbol{\beta}} Q = -2\mathbf{X}^\top\mathbf{y} + 2\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ yields the normal equations:

$$\mathbf{X}^\top\mathbf{X}\,\boldsymbol{\beta} = \mathbf{X}^\top\mathbf{y}.$$

Provided $\mathbf{X}$ has full column rank,[6] $\mathbf{X}^\top\mathbf{X}$ is strictly positive definite and thus invertible. The unique global minimizer is given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}. \tag{12}$$

## C.2 Proof of Two Sample Regression Equivalence

For convenience, we state the theorem formally before presenting the derivation.

**Theorem C.1**
Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be the OLS estimators given by (12) for the model $Y_i = \beta_0 + \beta_1 D_i + \varepsilon_i$, where $D_i = 1$ if observation $i$ is a weekend shopper and $D_i = 0$ if observation $i$ is a weekday shopper. The group sample sizes are defined as $n_1 = \sum_{i=1}^n D_i$ for weekend shoppers and $n_2 = \sum_{i=1}^n (1 - D_i)$ for weekday shoppers, with total sample size $n = n_1 + n_2$. Then:

1. $\hat{\beta}_0 = \bar{Y}_2$ and $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_2$.
2. $\text{RSS} = (n_1 - 1)S_1^2 + (n_2 - 1)S_2^2$, and the estimated error variance $\hat{\sigma}^2 = \text{RSS}/(n-2)$ is identical to the pooled sample variance $S_p^2$.
3. The OLS $t$ statistic for testing $H_0 : \beta_1 = 0$ is algebraically identical to the pooled two sample $t$ statistic.

*Proof.*

Recall that the response vector $\mathbf{y}$, design matrix $\mathbf{X}$, and parameter vector $\boldsymbol{\beta}$ are defined in matrix form as:

$$\mathbf{y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & D_1 \\ \vdots & \vdots \\ 1 & D_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

By construction, $\sum_{i=1}^n D_i = n_1$, $\sum_{i=1}^n (1 - D_i) = n_2$, and $D_i^2 = D_i$. Consequently, the matrix products in the normal equations:

$$\mathbf{X}^\top\mathbf{X}\,\boldsymbol{\beta} = \mathbf{X}^\top\mathbf{y}$$

evaluate to

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} \sum_{i=1}^n 1 & \sum_{i=1}^n D_i \\ \sum_{i=1}^n D_i & \sum_{i=1}^n D_i^2 \end{pmatrix} = \begin{pmatrix} n & n_1 \\ n_1 & n_1 \end{pmatrix},$$

and

$$\mathbf{X}^\top\mathbf{y} = \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n D_i Y_i \end{pmatrix} = \begin{pmatrix} n_1\bar{Y}_1 + n_2\bar{Y}_2 \\ n_1\bar{Y}_1 \end{pmatrix}.$$

[6] Full column rank requires that the columns of $\mathbf{X}$ are linearly independent.

The normal equations $\mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top\mathbf{y}$ expand to the system:

$$n\hat{\beta}_0 + n_1\hat{\beta}_1 = n_1\bar{Y}_1 + n_2\bar{Y}_2, \tag{13}$$

$$n_1\hat{\beta}_0 + n_1\hat{\beta}_1 = n_1\bar{Y}_1. \tag{14}$$

Assuming $n_1 > 0$, dividing (14) by $n_1$ yields

$$\hat{\beta}_0 + \hat{\beta}_1 = \bar{Y}_1 \implies \hat{\beta}_1 = \bar{Y}_1 - \hat{\beta}_0.$$

Substituting $\hat{\beta}_1$ into (13) gives

$$n\hat{\beta}_0 + n_1(\bar{Y}_1 - \hat{\beta}_0) = n_1\bar{Y}_1 + n_2\bar{Y}_2.$$

Simplifying the left hand side using $n - n_1 = n_2$:

$$(n - n_1)\hat{\beta}_0 + n_1\bar{Y}_1 = n_1\bar{Y}_1 + n_2\bar{Y}_2 \implies n_2\hat{\beta}_0 = n_2\bar{Y}_2.$$

Assuming $n_2 > 0$, we obtain

$$\hat{\beta}_0 = \bar{Y}_2 \qquad \text{and} \qquad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_2,$$

which proves the first assertion.

**Residual Sum of Squares** The fitted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 D_i$ evaluates to $\bar{Y}_1$ when $D_i = 1$ and $\bar{Y}_2$ when $D_i = 0$. Let the sample variances for the weekend and weekday groups be defined, respectively, as:[7]

$$S_1^2 = \frac{1}{n_1 - 1}\sum_{i:D_i=1}(Y_i - \bar{Y}_1)^2 \qquad \text{and} \qquad S_2^2 = \frac{1}{n_2 - 1}\sum_{i:D_i=0}(Y_i - \bar{Y}_2)^2.$$

The residual sum of squares partitions as:

$$\mathrm{RSS} = \sum_{i:D_i=1}(Y_i - \bar{Y}_1)^2 + \sum_{i:D_i=0}(Y_i - \bar{Y}_2)^2 = (n_1 - 1)S_1^2 + (n_2 - 1)S_2^2.$$

Since the model estimates $p = 2$ parameters, the residual degrees of freedom are $n - 2 = n_1 + n_2 - 2$. The estimated error variance is therefore:

$$\hat{\sigma}^2 = \frac{\mathrm{RSS}}{n - 2} = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} = S_p^2,$$

which matches the pooled sample variance estimator $S_p^2$.

**Standard Error and $t$ Statistic** The determinant of $\mathbf{X}^\top\mathbf{X}$ is:

$$\det(\mathbf{X}^\top\mathbf{X}) = nn_1 - n_1^2 = n_1(n - n_1) = n_1 n_2.$$

Inverting $\mathbf{X}^\top\mathbf{X}$ yields:

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{n_1 n_2}\begin{pmatrix} n_1 & -n_1 \\ -n_1 & n \end{pmatrix} = \begin{pmatrix} \frac{1}{n_2} & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \frac{n}{n_1 n_2} \end{pmatrix}.$$

[7] The denominators $n_1 - 1$ and $n_2 - 1$ are chosen so that $S_1^2$ and $S_2^2$ are unbiased estimators of the group variances, satisfying $\mathbb{E}[S_1^2] = \sigma_1^2$ and $\mathbb{E}[S_2^2] = \sigma_2^2$.

Under classical OLS assumptions, the full covariance matrix of $\hat{\boldsymbol{\beta}} = \begin{pmatrix}\hat{\beta}_0 \\ \hat{\beta}_1\end{pmatrix}$ is $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$, which expands entry by entry as:

$$\begin{pmatrix} \mathrm{Var}(\hat{\beta}_0) & \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) & \mathrm{Var}(\hat{\beta}_1) \end{pmatrix} = \sigma^2 \begin{pmatrix} \frac{1}{n_2} & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \frac{1}{n_1} + \frac{1}{n_2} \end{pmatrix}.$$

Extracting the lower right entry gives the sampling variance of $\hat{\beta}_1$:

$$\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right).$$

Replacing $\sigma^2$ with its unbiased estimator $\hat{\sigma}^2 = S_p^2$, the estimated variance is:

$$\widehat{\mathrm{Var}}(\hat{\beta}_1) = S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right).$$

Taking the square root yields the estimated standard error:

$$\widehat{\mathrm{SE}}(\hat{\beta}_1) = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

The resulting OLS $t$ statistic for testing $H_0 : \beta_1 = 0$ is:

$$t = \frac{\hat{\beta}_1 - 0}{\widehat{\mathrm{SE}}(\hat{\beta}_1)} = \frac{\bar{Y}_1 - \bar{Y}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

which is algebraically identical to the pooled two sample $t$ statistic (5) with $n_1 + n_2 - 2$ degrees of freedom. $\square$

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[1] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor. *An Introduction to Statistical Learning: with Applications in Python*. Springer Texts in Statistics. Springer, 2023.